Week 1 – Project Understanding & Planning

Title:

AI-powered campus assistant for navigation and information.

Abstract:

The project aim is about to create a bot to give the instructions or guiding students, staff, and new visitors across the university visiting .The system will integrate natural language processing (NLP) with campus maps to provide real-time navigation root, department information, and answers to asked questions. Using the tools like python and google maps and assistant will be available as a chatbot like direct python code out put or a website. This solution will enhance user experience, reduce confusion for new visitors , and streamline campus exploration.

Introduction:

Chanakya University campus are often complex, with multiple departments, hostels, and facilities like library , cafeteria, sports ground , food court, flag post, and large areas. New students, parents, and visitors face difficulty in locating specific buildings information quickly. Traditional notice websites are not always user-friendly for real-time guidance.

With advancements in artificial intelligence and conversational interfaces, chatbots integrated with navigation tool can provide interactive assistance. This project focuses on designing such a system tailored for campus use.

Problem Statement:

New students and visitors often face challenges in navigating the campus like where to go for to take admission , locating departments, or accessing FAQs efficiently. Existing systems like static maps or websites lack interactivity and personalization. There is a need for a smart assistant that combines natural language interaction with location-based services to improve campus accessibility.

Objectives:

- Develop an AI chatbot for campus navigation .
- Integrate campus layout/maps with Google Maps API for real-time guidance.
- Provide quick access to department details and campus resources.
- Ensure the system is scalable, user-friendly, and accessible on chatbot to get the information in it and free to use.
- Reduce dependency on manual inquiry/help desks.

Scope:

In-Scope:

- Navigation inside campus (departments, hostels, libraries, labs, offices).
- Department information (contacts, courses, services).
- Platform chatbot .

Requirements Collection:

- **Campus layout & maps** (digital blueprints).
- Departments and main visiting places
- **User groups:** Students, staff, parents, visitors.

Tools & Technology Finalization:

- **Programming Language:** Python.
- **APIs:** Google Maps API for navigation.